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Form factors in semileptonic decay $D \to Kl\bar{\nu}$ with Lattice QCD

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Abstract

Form factors in decay $D \to Kl\bar{\nu}$ are researched by using Lattice QCD with $2+1$-flavor domain wall fermion configurations. Valence quark action is Overlap fermion. The lattice size is $32^3 \times 64$ with an inverse lattice spacing $\frac{1}{a} = 2.383(9)$ GeV. Form factors defined by the scalar, vector and tensor current hadron matrix elements are computed.

Introduction

Form factors play a pivotal role in meson decay processes and are directly linked to the elements of the Cabibbo-Kobayashi-Maskawa (CKM) matrix, which serve as fundamental parameters of the Standard Model (SM) [1, 2].

To perform rigorous tests of the SM and search for potential new physics (NP), precise determinations of these parameters are indispensable. Achieving high precision requires a delicate balance between experimental measurements and theoretical predictions. LQCD is one of the primary tools for performing detailed investigations of these parameters.

Form factors derived from the scalar and vector currents provide a means of testing the SM [3, 4], whereas form factors associated with tensor currents are valuable for exploring scenarios beyond the SM and identifying signals of NP [5, 6].

Formulas and Lattice Setup

The form factors f_0, f_+ and f_T for the scalar, vector and tensor current respectively are defined as follows,

$$\langle K(p')|\bar{s}c|D(p)\rangle = \frac{m_D^2 - m_K^2}{m_c - m_s} f_0(q^2),$$

$$\langle K(p')|\bar{s}\gamma_\mu c|D(p)\rangle = f_+(q^2)\left[p_\mu + p'_\mu - \frac{m_D^2 - m_K^2}{q^2} q_\mu\right] + f_0(q^2)\frac{m_D^2 - m_K^2}{q^2} q_\mu,$$

$$\langle K(p')|\bar{s}\sigma_{\mu\nu} c|D(p)\rangle = \frac{2}{m_D + m_K}[p_\mu p'_\nu - p_\nu p'_\mu] f_T(q^2).$$

Those matrix elements are contained in the correlation functions,

$$C^{D\to K}(t, T; \vec{p}, \vec{p}') \propto \langle K(p')|\bar{s}\Gamma c|D(p)\rangle.$$

In this work, the gauge configurations were generated by the RBC/UKQCD collaborations [7]. Configuration ensemble and valence quark masses used in this work are shown in Tab. I and Tab. II, respectively.

am_l/am_s	m_π(MeV)	$N_{conf} \times N_{src}$
0.006/0.03	360	42×16

Table I. The configuration ensembles used in this work.

am_l	0.024, 0.0129, 0.00765
am_s	0.037, 0.040, 0.043, 0.046, 0.049
am_c	0.450, 0.492, 0.500, 0.550

Table II. Valence quark masses used in this work.

Preliminary Results

In this work, the least χ^2 method was employed to extract form factors from correlation functions. And renormalization constants are cite from Ref. [8] with value of $Z_V = 1.0789(10)$ for vector current and value of $Z_T = 1.157(11)$ in the $\overline{\text{MS}}$ scheme at 2 GeV for tensor current.

The form factors $f_0(q^2)$ and $f_+(q^2)$ were studied at four different q^2 values, while the form factor f_T was examined at three different q^2 values, with varying quark masses provided in Tab. II. And the relationship between the form factors, the transfer momentum, and the quark mass was investigated, with the results presented in Fig. I.

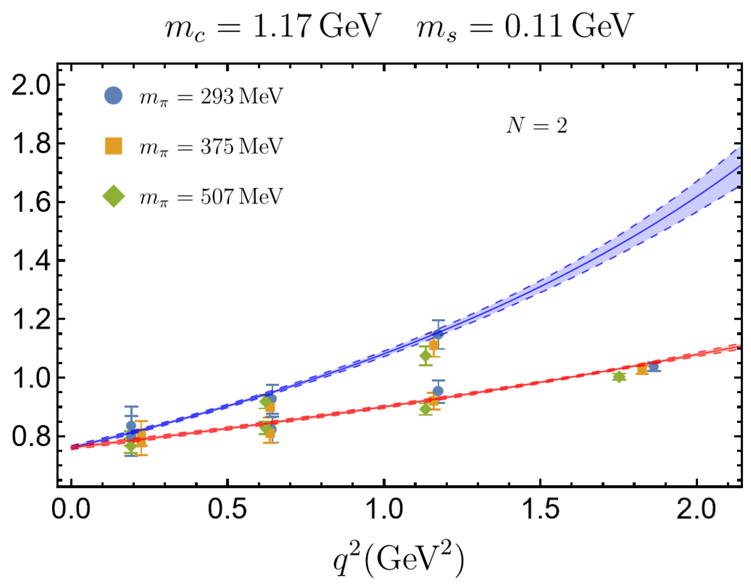


Figure. I The lattice results of form factor $f_0(q^2)$ and $f_+(q^2)$ based on four different q^2, and quark masses listed in Tab. II. The colored lines obtained by considering all masses points in Tab. II and error bars vary only with light quark mass, while c and s quark masses are fixed.

Similarly, the behavior of form factor $f_{\text{T}}(q^2)$ was also investigated, with the corresponding results shown in Fig. II.

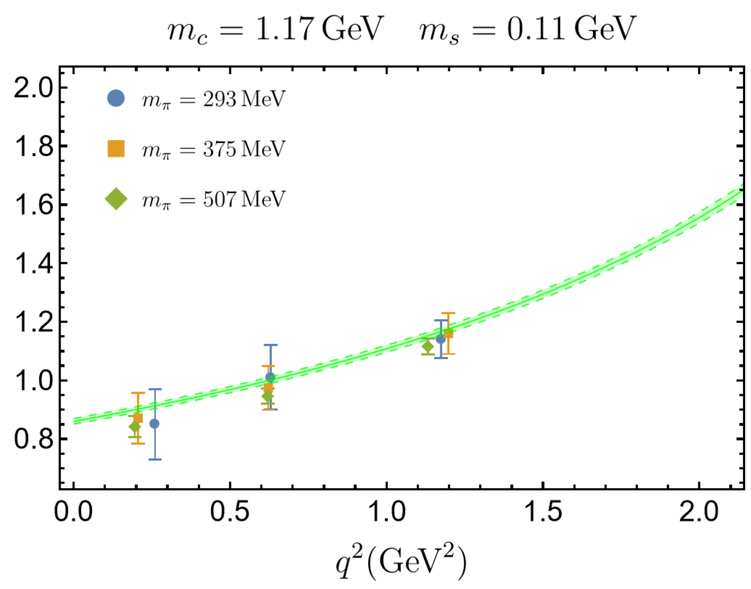


Figure. II The lattice results of form factor $f_{\text{T}}(q^2)$ based on three different q^2, and quark masses in Tab. II.

Outlook

The lattice results will be extrapolated or interpolated to the physical point.

To reach the continuum limit and estimate systematic uncertainties, more configuration ensembles will be required.

References

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